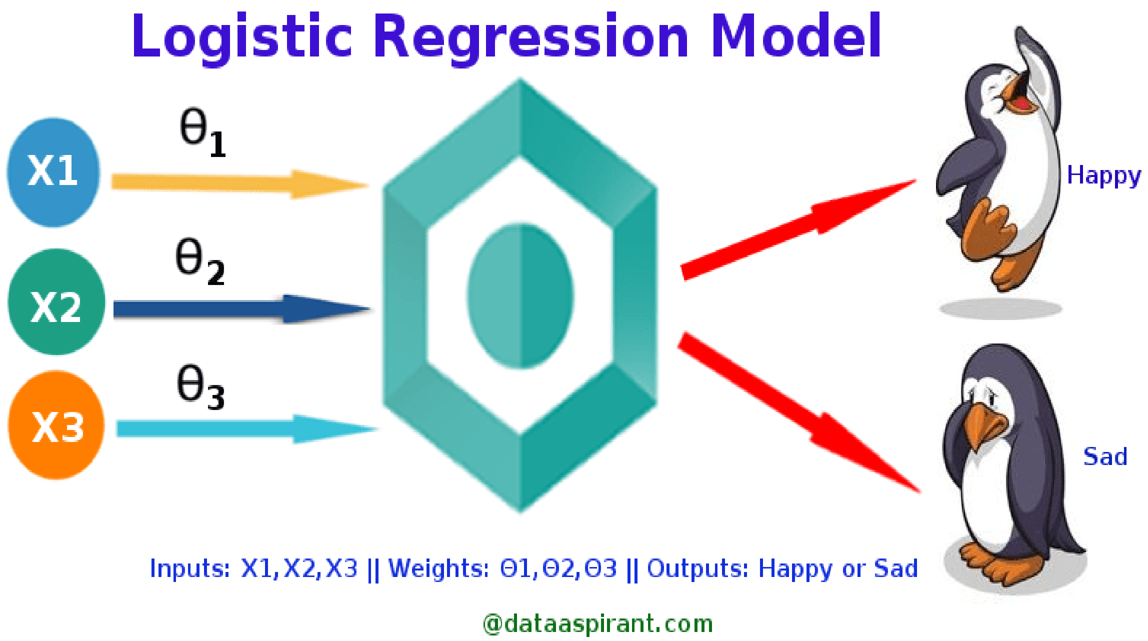
# Logistic Regression

# What is logistic regression?

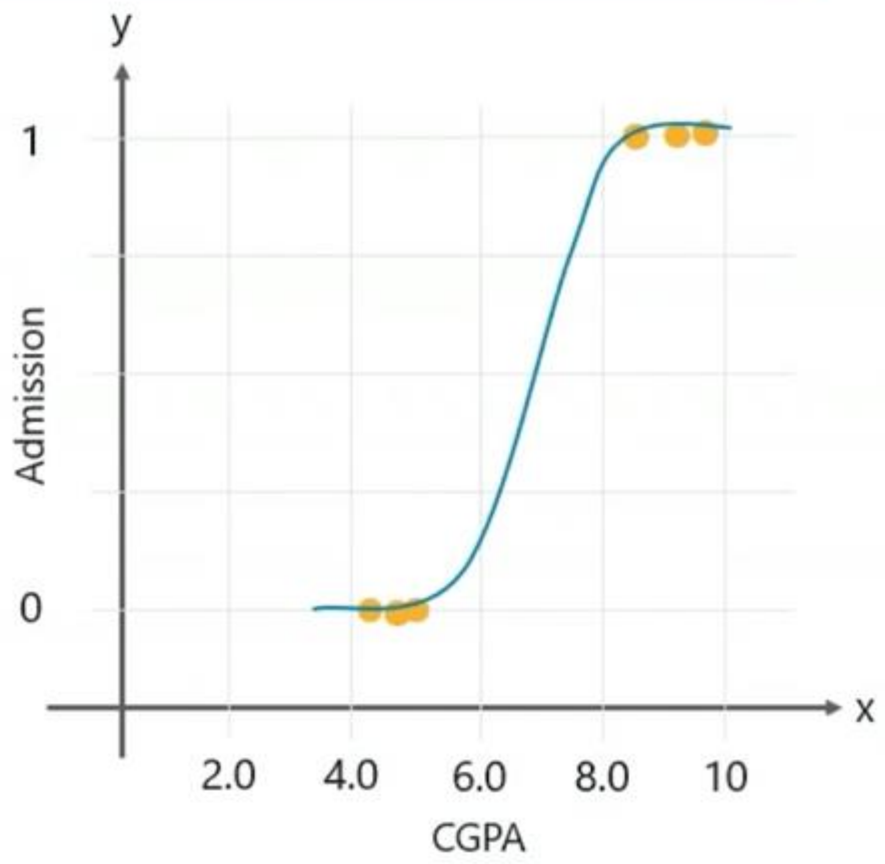
- This type of statistical model (also known as *logit model*) is often used for classification and predictive analytics.
- **Logistic Regression is used when the dependent variable(target) is categorical.**
- It estimates the probability of an event occurring, such as voted or didn’t vote (1 or 0, YES or No), based on a given dataset of independent variables.
- For example,
  - To predict whether an email is spam (1) or (0)
  - Whether the tumor is malignant (1) or not (0)
- Since the outcome is a probability, the dependent variable is bounded between 0 and 1.
- The parameters of a logistic regression are most commonly estimated by maximum-likelihood estimation (MLE).



# Logistic Regression Use Case

To predict if a student will get admitted to a school based on his CGPA.

| Admission | CGPA |
|---|---|
| 0 | 4.2 |
| 0 | 5.1 |
| 0 | 5.5 |
| 1 | 8.2 |
| 1 | 9.0 |
| 1 | 9.1 |



# How does Logistic regression work?

- The logistic regression model transforms the [linear regression](linear regression) function continuous value output into categorical value output using a sigmoid function, which maps any real-valued set of independent variables input into a value between 0 and 1.
- This function is known as the logistic function.
- Let the independent input features be:

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ x_{21} & \cdots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{bmatrix}$$

- Let the dependent variable is Y, having only binary value i.e. 0 or 1.

$$Y = \begin{cases} 0 & \text{if } Class\ 1 \\ 1 & \text{if } Class\ 2 \end{cases}$$

- Apply the multi-linear function to the input variables X.

$$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_nx_n$$

- Here xi is the ith observation of X, b*i* is the weights or Coefficient, and $b_0$ is the bias term also known as intercept. This can be represented as the dot product of weight and bias.

# Logistic Regression Equation:

In Logistic Regression y can be between 0 and 1 only, so we use the sigmoid function where the input will be y and we find the probability between 0 and 1. i.e. predicted y.

$$z = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \cdots + b_n x_n$$

$$P(y=1) = \frac{1}{1+e^{-z}}$$

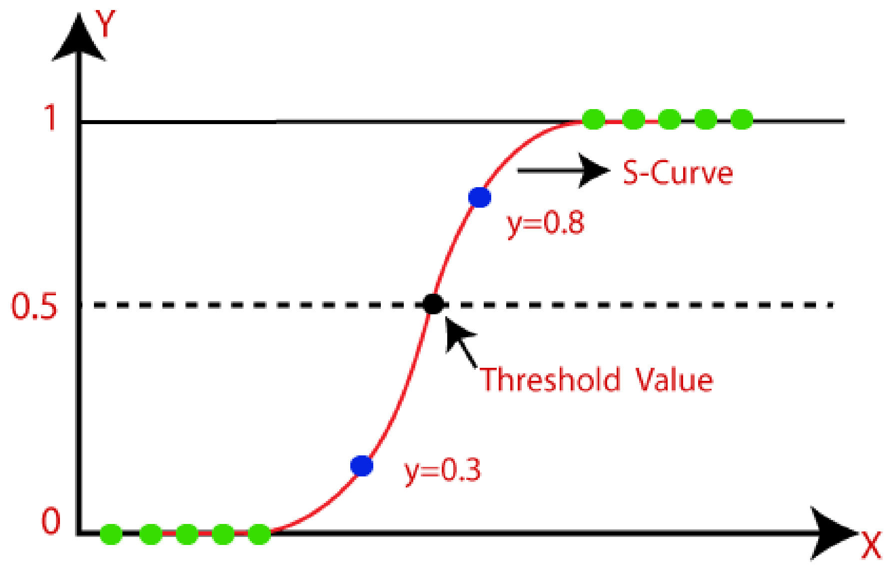
**Logit (Log-Odds) Form**

Another common way to write logistic regression is:

$$\ln\left(\frac{P(y=1)}{1-P(y=1)}\right) = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \cdots + b_n x_n$$

## Logistic Function (Sigmoid Function):

- The logistic regression model transforms the [linear regression] function continuous value output into categorical value output using a sigmoid function, which maps any real-valued set of independent variables input into a value between 0 and 1.
- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1.
- The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.



# ESTIMATED REGRESSION EQUATION

The natural logarithm of the odds ratio is equivalent to a *linear* function of the independent variables. The antilog of the logit function allows us to find the estimated regression equation.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 \quad \text{antilog} \quad \frac{p}{1-p} = e^{\beta_0+\beta_1 x_1}$$

$$p = e^{\beta_0+\beta_1 x_1}(1-p)$$

$$p = e^{\beta_0+\beta_1 x_1} - e^{\beta_0+\beta_1 x_1} * p$$

$$p + e^{\beta_0+\beta_1 x_1} * p = e^{\beta_0+\beta_1 x_1}$$

$$p(1 + e^{\beta_0+\beta_1 x_1}) = e^{\beta_0+\beta_1 x_1}$$

$$\hat{p} = \frac{e^{\beta_0+\beta_1 x_1}}{1 + e^{\beta_0+\beta_1 x_1}}$$

Estimated Regression Equation

- we need range between -[infinity] to +[infinity], then take logarithm of the equation it will become:

$$\log\left[\frac{y}{1-y}\right] = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \cdots + b_n x_n$$

- In logistic regression, a logit transformation is applied on the odds—that is, the probability of success divided by the probability of failure.
- This is also commonly known as the log odds, or the natural logarithm of odds (The odds ratio is the ratio of odds of an event A in the presence of the event B and the odds of event A in the absence of event B).
- P is the probability that event Y occurs. P(Y=1)
- P/(1-P) is the odds ratio

- **Logistic Regression can be used for binary classification or multi-class classification.**
  - *Binary classification is when we have two possible outcomes like a person is infected with COVID-19 or is not infected with COVID-19.*
  - *In multi-class classification, we have multiple outcomes like the person may have the flu or an allergy, or cold or COVID-19.*
- **Assumptions for Logistic Regression**
  - No outliers in the data. (An outlier can be identified by analyzing the independent variables)
  - No correlation (multi-collinearity) between the independent variables.

# Linear Regression Vs Logistic Regression

| | Linear Regression | Logistic Regression |
|---|---|---|
| 1 Definition | To predict a continuous dependent variable based on values of independent variables | *To predict a categorical dependent variable based on values of independent variables* |
| 2 Variable Type | Continuous dependent variable | Categorical dependent variable |
| 3 Estimation method | Least square estimation | Maximum like-hood estimation |
| 4 Equation | $Y = b_0 + b_1 x + e$ | $\log\left(\frac{Y}{1-Y}\right) = C + B1X1 + B2X2 + ....$ |
| 5 Best fit line | Straight line | Curve |
| 6 Relationship between DV & IV | Linear relationship between the dependent and independent variable | Linear relationship is not mandatory |
| 7 Output | Predicted integer value | Predicted binary value (0 or 1) |
| 8 Applications | Business domain, forecasting sales | Classification problems, cybersecurity, image processing |

# Reference

- Logistic Regression: https://www.youtube.com/watch?v=yIYKR4sgzI8
- Maximum Likelihood Estimation: https://www.youtube.com/watch?v=XepXtl9YKwc